

Bayes' Rule & Example

Video Lesson

Duration: 00:20:41

Slide Deck

M11S04 (<https://canvas.tamu.edu/courses/430127/files/92570690?wrap=1>) 

Lesson Transcript

In this lecture, we will look at the Bayes' Rule and some examples. The Bayes' Rule plays a very important role in modern probabilistic AI and also in machine learning.

Bayes' Rule

- From $P(A|B) = \frac{P(A \cap B)}{P(B)}$ and $P(B|A) = \frac{P(A \cap B)}{P(A)}$, we get

$$P(A|B)P(B) = P(B|A)P(A)$$

and in turn from which we get the **Bayes' Rule**:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

The Bayes' Rule is pretty simple. Let us look at this formula at the bottom of the page. The probability of B given A is the probability of A given B times the probability of B divided by the probability of A .

Using the Bayes' Rule, we can flip this conditional probability, the probability of B given A , into the probability of A given B by applying some of these other probabilities and multiplying and dividing.

The main reason for doing this is, in many cases, estimating this conditional probability is easier than directly estimating that probability. And we look at this in the following example.

The Bayes' Rule can be directly derived from the definition of conditional probability. The probability of A given B is the probability of A and B divided by the probability of B . The Probability of B given A is the probability of A and B divided by the probability of A .

Bayes' Rule

$$P(A, B) = P(B, A)$$

- From $P(A|B) = \frac{P(A \cap B)}{P(B)}$ and $P(B|A) = \frac{P(A \cap B)}{P(A)}$, we get

$$P(A|B)P(B) = P(B|A)P(A)$$

and in turn from which we get the **Bayes' Rule**:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Note that for joint probability, the order does not matter. That is, the probability of A and B equals the probability of B and A . So, that is what we use here.

Bayes' Rule

- From $P(A|B) = \frac{P(A \cap B)}{P(B)}$ and $P(B|A) = \frac{P(A \cap B)}{P(A)}$, we get

$$P(A|B)P(B) = P(B|A)P(A)$$

and in turn from which we get the **Bayes' Rule**:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

From these two equalities, let us multiply both sides with $P(B)$ for that case, and for this case, let us multiply both sides with $P(A)$.

What happens next is these two cancel out, leaving only $P(A \cap B)$ on the right-hand side for that. And $P(A|B)$ multiplied by $P(B)$ on the left-hand side. We can see that this and this, the two are the same. Same here, 1 over $P(A)$ and $P(A)$. They cancel out, leaving $P(A \cap B)$ on the right-hand side and the probability of B given A times $P(A)$ on the left-hand side.

And since these two are equal, these two also should be equal.

And when we write that down, then we get this here.

Bayes' Rule

- From $P(A|B) = \frac{P(A \cap B)}{P(B)}$ and $P(B|A) = \frac{P(A \cap B)}{P(A)}$, we get

$$\frac{P(A|B)P(B)}{P(A)} = \frac{P(B|A)P(A)}{P(A)}$$

and in turn from which we get the **Bayes' Rule**:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Next, we divide both sides with $P(A)$. Then these two $P(A)$ cancel out, and we have $P(B | A)$ on the right-hand side. And $P(A | B)$, $P(B)$ over $P(A)$ on the left-hand side. These are identical, and these are identical. Hence we have the Bayes' Rule.

Extended Bayes' Rule

$$P(Y|X, E) = \frac{P(X|Y, E)P(Y|E)}{P(X|E)}$$

Derivation:

$$\begin{aligned} P(Y|X, E) &= \frac{P(Y, X, E)}{P(X, E)} \\ &= \frac{P(X, Y, E)}{P(X, E)} \\ &= \frac{P(X|Y, E)P(Y, E)}{P(X, E)} \\ &= \frac{P(X|Y, E)P(Y|E)P(E)}{P(X|E)P(E)} \\ &= \frac{P(X|Y, E)P(Y|E)}{P(X|E)} \end{aligned} \tag{1}$$

Note: $P(A|B, C) = P(A|(\underbrace{B, C}))$, and using $P(A|B) = P(A, B)/P(B)$ and $P(A, B) = P(A|B)P(B)$.

Now we can extend this Bayes' Rule to the Extended Bayes' Rule, which looks like this probability

of Y given X , E equals the probability of X given Y , E times probability of Y given E , over the probability of X given E .

Extended Bayes' Rule

$$P(Y|X, E) = \frac{P(X|Y, E)P(Y|E)}{P(X|E)}$$

$$\rightarrow P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

(1)

Derivation:

$$\begin{aligned} P(Y|X, E) &= \frac{P(Y, X, E)}{P(X, E)} \\ &= \frac{P(X, Y, E)}{P(X, E)} \\ &= \frac{P(X|Y, E)P(Y, E)}{P(X, E)} \\ &= \frac{P(X|Y, E)P(Y|E)P(E)}{P(X|E)P(E)} \\ &= \frac{P(X|Y, E)P(Y|E)}{P(X|E)} \end{aligned}$$

Note: $P(A|B, C) = P(A|(B, C))$, and using $P(A|B) = P(A, B)/P(B)$ and $P(A, B) = P(A|B)P(B)$.

One way of looking at this is to eliminate this extra part E . Then what remains is basically the same Bayes' Rule.

When we plug it back in, the sort and the variable E to the right side of the bar for the conditional probability, then this becomes probability Y given X , E . Like that. This becomes the probability of X given Y and E . It is that one. Here, there is a prior probability, probability Y . Now because we want to add this to the right-hand side of the bar, this becomes the probability of Y given E .

Finally, this becomes P of X to the probability of X given E .

Extended Bayes' Rule

$$P(Y|X, E) = \frac{P(X|Y, E)P(Y|E)}{P(X|E)}$$

Derivation:

$$\begin{aligned}
 P(Y|X, E) &= \frac{P(Y, X, E)}{P(X, E)} \\
 &= \frac{P(X, Y, E)}{P(X, E)} \\
 &= \frac{P(X|Y, E)P(Y, E)}{P(X, E)} \\
 &= \frac{P(X|Y, E)P(Y|E)P(E)}{P(X|E)P(E)} \\
 &= \frac{P(X|Y, E)P(Y|E)}{P(X|E)}
 \end{aligned} \tag{1}$$

Note: $P(A|B, C) = P(A|(\underbrace{B, C}))$, and using $P(A|B) = P(A, B)/P(B)$ and $P(A, B) = P(A|B)P(B)$.

Here is the derivation. First, we want to emphasize that when we see probability Y given X, E . The X and E on the right-hand side of that bar move as one unit, as shown here. When we have the probability of A given B, C , we should always group these together like that. When we apply the Bayes' Rule to that, then we get something.

But, for now, let us just use the definition of conditional probability. The conditional probability of Y given X, E is the joint probability divided by this part, the probability of X, E . Here, we have the probability of Y and X and E , given the probability of X and E .

Then we just arranged the ordering of these random variables to go from the first line to the second line, and then we used the definition of conditional probability backward.

That is illustrated below. The probability of A and B is the probability of A given B times the probability of B . So, if we divide both sides with the probability of B , then that is the definition of the probability of A and B . Thus, we went from the probability of X, Y, E to the probability of X given Y, E multiplied by the probability of Y and E . We only changed that numerator part.

Then the denominator part carries over. Then we apply the same trick here. So, this becomes Y given E times $P(E)$. Then we apply the same trick here, P of X and E becomes P of X given E , $P(E)$. Finally, when we cancel out $P(E)$, then we finally have the right-hand side of this original equality.

Example: Application of Bayes Rule

After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease, and that the test is 99% accurate (i.e. the probability of testing positive given that you have the disease is 0.99, as is the probability of testing negative given you don't have the disease). The good news is that this is a rare disease, striking only 1 in 10,000 people. Why is it good news that the disease is rare? What are the chances that you actually have the disease?
 T : tested positive, $\neg T$: tested negative, D : have disease, $\neg D$: clean.

Here is an example application of the Bayes' Rule. Let us consider this situation. After our yearly checkup, the doctor has bad news and good news. The bad news is that we tested positive for a very serious disease and that this test is 99% accurate. The probability of testing positive if we have the disease is 0.99, and the probability of testing negative when we do not have the disease is 0.99.

There are two pieces of news here. First is a deadly disease. And the test is very accurate, it seems.

Of course, by this time, we are curious about the good news.

The doctor tells us that the good news is that this is a very rare disease, with the probability of 1 out of 10,000 people contracting this disease.

But now we are scratching our heads. Because we do not know what is good about this, it does not seem to help.

Example: Application of Bayes Rule

After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease, and that the test is 99% accurate (i.e. the probability of testing positive given that you have the disease is 0.99, as is the probability of testing negative given you don't have the disease). The good news is that this is a rare disease, striking only 1 in 10,000 people. Why is it good news that the disease is rare? What are the chances that you actually have the disease?

T : tested positive, $\neg T$: tested negative, D : have disease, $\neg D$: clean.

$$P(T|D) = 0.99$$

$$P(D|T) = ?$$

Here the most important thing that we need to realize is that this 0.99 is basically this conditional probability. What is the probability of testing positive given a disease? What we really want to know is, what is the probability that we have the disease, given that we tested positive? Would this be about the same, or would it be different?

We do not know yet, and it looks like we can use Bayes' Rule to compute this using that. Because T and D are boolean random variables, we can just write T for testing positive, $\neg T$ for testing negative, D for having disease, and $\neg D$ for being clean.

Solution: Good News and Bad News

These are given:

$$\begin{aligned}P(T|D) &= 0.99 \\P(\neg T|\neg D) &= 0.99 \\P(D) &= \frac{1}{10,000} = 0.0001\end{aligned}$$

We want to calculate the probability that you have the disease given a positive test result:

$$P(D|T)$$

We can use Bayes' rule to derive this probability.

From the information given in the previous slide, we have these values readily available.

What is the probability of testing positive, given the disease? That is the accuracy of the test, which is 0.99. What is the probability of a negative test result given non-diseased? That is again 0.99. That is, again, the test accuracy.

The probability of disease is, as the doctor told us, 1 over 10,000. That is 0.0001.

Again, what we are really interested in is the probability of D given T .

Solution: Good News and Bad News

These are given:

$$\begin{aligned}P(T|D) &= 0.99 \\P(\neg T|\neg D) &= 0.99 \\P(D) &= \frac{1}{10,000} = 0.0001\end{aligned}$$

Guess

We want to calculate the probability that you have the disease given a positive test result:

$$P(D|T)$$



We can use Bayes' rule to derive this probability.

① ~ 0.9 ?
② ~ 0.1
③ ~ 0.01

We can try guessing what this would be. Would it be something like 0.9? Would it be something like 0.1? Or would it be something like 0.01?

We will find out later on that our intuition can be very wrong.

Solution: Good News and Bad News (cont'd)

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)}$$

- $P(T|D) = 0.99$, $P(\neg T|\neg D)$, and $P(D) = 0.0001$ are given.
- From these, we can get $P(\neg T|D) = 0.01$, $P(T|\neg D) = 0.01$, and $P(\neg D) = 0.9999$.

Since $P(T|D)$ and $P(D)$ are given, we only need to calculate $P(T)$.

To compute this posterior probability, we simply use Bayes' Rule. The probability of D given T is the probability of T given D times $P(D)$ over $P(T)$.

Solution: Good News and Bad News (cont'd)

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)}$$

- $P(T|D) = 0.99$, $P(\neg T|\neg D)$, and $P(D) = 0.0001$ are given.
- From these, we can get $P(\neg T|D) = 0.01$, $P(T|\neg D) = 0.01$, and $P(\neg D) = 0.9999$.

Since $P(T|D)$ and $P(D)$ are given, we only need to calculate $P(T)$.

On the right-hand side, these two values are already given in the description. Where is the test accuracy, which is 0.99? Then what is the prior probability of the disease, which is 0.0001? But how do we compute this probability of testing positive?

Solution: Good News and Bad News (cont'd)

$\Sigma = 1.0$

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)}$$

- $P(T|D) = 0.99$, $P(\neg T|\neg D)$, and $P(D) = 0.0001$ are given.
- From these, we can get $P(\neg T|D) = 0.01$, $P(T|\neg D) = 0.01$, and $P(\neg D) = 0.9999$.

Since $P(T|D)$ and $P(D)$ are given, we only need to calculate $P(T)$.

$\Sigma = 1.0$

From these three given values, we can also compute these three values directly by realizing that the sum of these two would be 1.0. We can just subtract this from 1.0 to get 0.01 for the probability of not testing positive, given the disease. The same for this. This is known. This is known. This is not known.

We subtract this value, which is 0.99, from 1.0 to get 0.01 and the same for the prior probability.

We need these values to compute $P(T)$.

Solution: Good News and Bad News (cont'd)

Observation 1: $P(T) = P(T|D)P(D) + P(T|\neg D)P(\neg D)$.

Thus, $P(T) = 0.99 \times 0.0001 + 0.01 \times 0.9999 = 0.010098$, and with this,

$$P(D|T) = \frac{0.99 \times 0.0001}{0.010098} = 0.0098,$$

which is slightly less than 1%.

Exercise: how accurate should the test be so that $P(D|T)$ is greater than 0.95 (i.e. 95%)?

This $P(T)$ can be computed as $P(T|D)$, $P(D)$ plus P of T , given $\neg D$ times P of $\neg D$. These values are all available from the derivation in the previous slide. When we plug in these values, the P of T , given D , is 0.99. P of D is 0.0001. Then the P of T given $\neg D$ is 0.01. P of $\neg D$ is 0.9999.

When we add this all up, then we get 0.010098 for $P(T)$. If we plug in the known values P of T given D of 0.99 and P of D of 0.0001 in the numerator and plug in this value, then surprisingly, this

posterior probability comes to 0.0098. So, it is even less than 1%.

By the way, this formula here comes from $P(T)$ equals P of T and D plus P of T and $\neg D$. We have seen this technique in one of the earlier slides, and we can use the definition of conditional probability to go from here to this product. We will look at that in more detail in the following slide.

Solution: Good News and Bad News (cont'd)

Observation ¹: $P(T) = P(T|D)P(D) + P(T|\neg D)P(\neg D)$.

Thus, $P(T) = 0.99 \times 0.0001 + 0.01 \times 0.9999 = 0.010098$, and with this,

$$P(D|T) = \frac{0.99 \times 0.0001}{0.010098} = 0.0098,$$



which is slightly less than 1%.

$$P(T|D) = 0.99$$

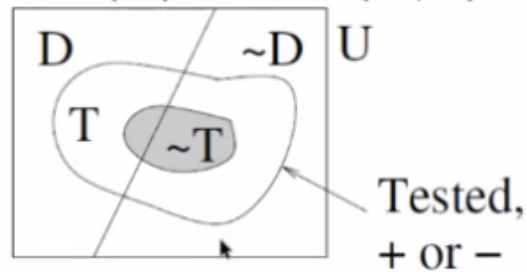
Exercise: how accurate should the test be so that $P(D|T)$ is greater than 0.95 (i.e. 95%)?

Now we see why the doctor mentioned that the good news is that this disease is very rare.

Compared to this likelihood, the probability P of T given D , which was 0.99. There is the accuracy of the test. The probability that we do have this disease given this test positive result is only about 1%, 0.0098. These are very different, and it is highly counterintuitive.

With this kind of formula, we can ask questions. For example, what should the test accuracy be so that this accuracy is over 0.95?

Calculating $P(T)$ given $P(T|D)$ and $P(D)$



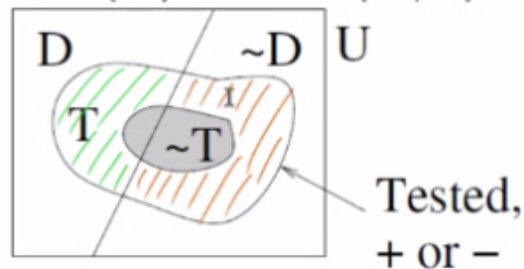
$$\begin{aligned} P(T) &= P(T \wedge D) + P(T \wedge \neg D) \\ &= P(T|D)P(D) + P(T|\neg D)P(\neg D) \end{aligned}$$

- $\{D\} \cup \{\neg D\}$ completely account for the whole population, but $\{T\} \cup \{\neg T\}$ does not cover the whole population (because you **did not test everyone!**).

Here is an illustration for a better understanding of computing this probability, $P(T)$. Let us say U is the entire population. If we look at this entire population, everyone is either diseased or not diseased, so they will somehow divide the entire set, and there will be no leftovers. However, only a small fraction of the entire population would have been tested, and of those that were tested, some tested positive, and some tested negative.

To compute $P(T)$, we must find the area for that donut shape, excluding that.

Calculating $P(T)$ given $P(T|D)$ and $P(D)$



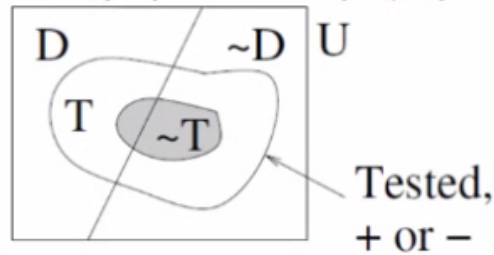
$$\begin{aligned} P(T) &= P(T \wedge D) + P(T \wedge \neg D) \\ &= P(T|D)P(D) + P(T|\neg D)P(\neg D) \end{aligned}$$

- $\{D\} \cup \{\neg D\}$ completely account for the whole population, but $\{T\} \cup \{\neg T\}$ does not cover the whole population (because you **did not test everyone!**).

The probability of T and D corresponds to this area. The probability of T and $\neg D$ corresponds to this area. Here we have T , and then this region is $\neg T$. Here again, these belong to T , and that

whole region is D diseased.

Calculating $P(T)$ given $P(T|D)$ and $P(D)$



$$\begin{aligned} P(T) &= P(T \cap D) + P(T \cap \neg D) \\ &= P(T|D)P(D) + P(T|\neg D)P(\neg D) \end{aligned}$$

$$P(A \cap B) = P(A|B)P(B)$$

$$\frac{P(A \cap B)}{P(B)} = P(A|B)$$

- $\{D\} \cup \{\neg D\}$ completely account for the whole population, but $\{T\} \cup \{\neg T\}$ does not cover the whole population (because you **did not test everyone!**).

For this line going to that line, we again use this definition of conditional probability multiplied by both sides with $P(B)$. It will give us this P of A and B equals P of A given B , $P(B)$. That is exactly what we did here, and that is the formula that we used earlier in the previous slide.

Calculating $P(T)$ given $P(T|D)$ and $P(D)$

Another way of deriving $P(T) = P(T|D)P(D) + P(T|\neg D)P(\neg D)$:
From

$$\begin{aligned} P(D|T) &= \frac{P(T|D)P(D)}{P(T)} \\ P(\neg D|T) &= \frac{P(T|\neg D)P(\neg D)}{P(T)} \end{aligned}$$

and from $P(D|T) + P(\neg D|T) = 1$,

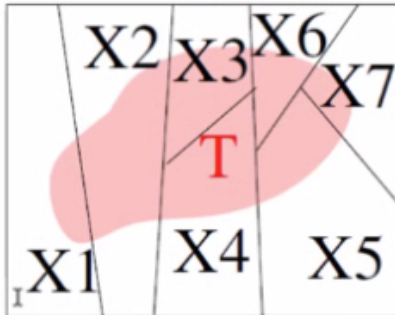
$$\begin{aligned} 1 &= \frac{P(T|D)P(D)}{P(T)} + \frac{P(T|\neg D)P(\neg D)}{P(T)}, \text{ thus} \\ P(T) &= P(T|D)P(D) + P(T|\neg D)P(\neg D) \end{aligned}$$

Here is another way of deriving $P(T)$. Let us look at P of D given T and P of $\neg D$ given T . Then, because it can be either D or $\neg D$ given T , when we add these two up, we will get 1.

In each of these, we can apply the Bayes' Rule to get this and that for these two. This here, plus that there, equals 1.

When we plug in this value in place of that and this in place of that here. We so far have established this: 1 equals P of T given D , P of D over $P(T)$ plus P of T , given $\neg D$, P of $\neg D$ over $P(T)$. Now, we can simply multiply both sides with P of T . Then we get the desired formula.

Calculating $P(T)$: General Case



More Generally, if $\left[\sum_{x \in \{x_1, x_2, \dots, x_n\}} P(X = x) \right] = 1$ and events $X = x_m$ and $X = x_n$ are disjoint for all $m \neq n$,

$$P(T) = \sum_{x \in \{x_1, x_2, \dots, x_n\}} P(T | \underbrace{X = x}) P(X = x)$$

We can generalize this to the case where we have a random variable X that can take on seven different values, as shown here. Let us say it covers the entire population. Here we have T . Then the probability of T is computed as the sum of P of T given X equals X_1 times P of X equals X_1 , plus P of T given X equals X_2 multiplied by P of X equals X_2 , and so on over all possible X values.

What's The Big Deal?

- $P(T|D)$ may be easier to obtain: you can run the test on a small pool of known patients (say 100) at a hospital.
- $P(D|T)$ is much harder to obtain directly. Since the test makes 1 mistake out of 100 tests, if you run the test on 10,000 people, you'll get 100 false-positives, and one genuine patient who tests positive (consider that $P(T) = 0.010098$). So, just to get about 100 people testing positive, you have to run the tests on 10,000 people.
- $P(D)$ serves as a **prior** in this case. In many cases, the prior represents subjective **belief** of the person calculating the probability in case $P(D)$ is not directly measurable.

What is the big deal here? Why is using Bayes' Rule more convenient or more effective? First, let us consider this likelihood: the probability of testing positive given disease. We can quickly generate enough samples for this or estimate the probability for this by just going to a hospital where there is a large population of people with that specific disease.

Whereas if we try to directly estimate this posterior probability, the probability of disease given test, then we will have to go out on the street and test people. But the problem is that this is a very rare disease. If this test was a hundred percent accurate after testing 10,000 people, we get one diseased person. That is 1 sample after 10,000 people.

If this test is 99% accurate, then there will be some false positives. Out of 10,000, 1% would turn positive. That means we will get about a hundred people with a positive result. And many of that would be false positives.

Just to get these hundred positive results and find a particular diseased person, we have to go through 10,000 people. It can be very, very time-consuming and inefficient.

Another interesting thing about this or the use of Bayes' Rule is the probability P of D , which is the prior probability.

When this D changes or when this probability of D changes, the probability of D given T will change. If we want to directly compute that, then depending on the particular probability, this could be, for example, a seasonal thing. Then we have to change our estimate every time. Whereas if we use the Bayes' Rule, and if we know this prior probability, depending on a particular season, then we can just plug in this value where this particular probability testing positive given having a disease is more or less a fixed quantity because it is based on known biochemical reactions.

So, instead of having to estimate this every time over and over again when there is any change in the prior probability, we can just plug in this prior probability and multiply that with the likelihood to

compute this directly.

One other interesting aspect about the Bayes Rule is that this prior, in many cases, is actually hard to estimate, and in this case, we can put in some kind of subjective belief in place of the known experimental value.

So far, we looked at Bayes' rule and its applications. In the following lecture, we will look at the situation when multiple pieces of evidence come in and how to update this kind of Bayesian estimate.

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